Spatio-Temporal Traffic Speed Forecasting on METR-LA: From Engineered Features to Graph Neural Networks

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***Abstract*—Placeholder abstract.**

***Index Terms*—traffic forecasting, spatio-temporal data, graph neural networks, Graph WaveNet, gradient boosting, feature engineering, METR-LA**

I. Introduction

Accurate short-term traffic forecasts underpin route guidance, ramp metering, incident response and travel-time estimation in Intelligent Transportation Systems [1]. Forecasting freeway speed is hard for four reasons that are all visible in real sensor data: the dynamics are strongly *non-linear* near the onset of congestion; the series are *seasonal* (daily and weekly demand cycles); congestion *propagates spatially* along the road network; and loop detectors frequently *fail*, producing missing readings.

Research has moved from univariate statistical models (ARIMA [2], [3]) through feature-based machine learning [4] and recurrent networks [5] to spatio-temporal graph neural networks (GNNs) that treat the sensor network as a graph [6]–[8]. Published comparisons, however, often pit sophisticated GNNs against feature-poor classical baselines, and rarely test whether accuracy differences are statistically significant.

This paper studies 1-hour-ahead speed forecasting on the METR-LA benchmark [6] and makes the following contributions:

1) A data-quality and exploratory analysis that quantifies missingness, seasonality, stationarity and spatial dependence. It links each finding to a modelling decision.
2) A leakage-tested feature set (34 features in six groups) that includes a weekly profile/anomaly decomposition and upstream/downstream graph aggregates. Its value is measured by cumulative ablation and permutation importance.
3) A controlled ladder of six forecasters (Historical Average, Persistence, Ridge, gradient boosting, LSTM and Graph WaveNet [8]) evaluated under the standard protocol. We compare against published results and report Diebold–Mariano tests [9] and block-bootstrap confidence intervals.
4) An ablation of Graph WaveNet's spatial module (none, road graph, adaptive graph, both) under an equal budget, plus error analyses by horizon, time of day, sensor and missing input.

All code, tests and results are released in the project repository.

II. Related Work

Statistical models. Ahmed and Cook [2] first applied Box–Jenkins ARIMA to freeway data, and Williams and Hoel [3] justified seasonal ARIMA by showing that weekly-differenced traffic is stationary. These models are univariate and linear, and they degrade quickly with horizon: on METR-LA, ARIMA's MAE rises from 3.99 to 6.90 mph between 15 and 60 min [6].

Feature-based machine learning. Non-parametric regression (k-NN) was shown to be competitive and robust [4]. Tree ensembles [10] and gradient boosting [11], [12] model nonlinear interactions but depend heavily on feature engineering. In the DCRNN benchmark, SVR on raw lags reached only 6.72 mph at 60 min [6].

Deep sequence models. Stacked autoencoders [13] and LSTMs [5], [14] learn temporal representations directly. Encoder–decoder (FC-)LSTMs [15] predict a whole horizon and reach 4.37 mph at 60 min on METR-LA [6], but they ignore road topology.

Spatio-temporal GNNs. Spectral [16] and first-order [17] graph convolutions made it possible to exploit topology. DCRNN [6] models traffic as a bidirectional diffusion process on a directed distance graph inside a GRU seq2seq. STGCN [7] replaces recurrence with gated temporal convolutions, and ASTGCN [18] adds attention. Graph WaveNet [8] combines diffusion convolution with dilated causal convolutions [19] and a *self-adaptive* adjacency learned from node embeddings, showing that the predefined graph is incomplete. Later work learns the graph entirely (MTGNN [20], AGCRN [21]), uses attention (GMAN [22], STAEformer [23]), or shows that simple spatial and temporal identity embeddings recover much of the gain (STID [24]). Jiang and Luo [25] survey the area.

TABLE I
DATASET SUMMARY (COMPUTED BY SCRIPTS/RUN_EDA.PY).

Sensors / time steps	207 / 34,272 (119 days, 5 min)
Missing readings (value = 0)	8.11 % (per sensor 6.3–20.1 %)
Observed speed mean ± std	58.5 ± 13.0 mph (median 63.2)
Readings < 40 mph (congested)	10.9 %
Skewness	−1.99
Graph non-zeros (incl. self-loops)	1,722 (1,515 directed edges)
One-way edges	73.3 %
Split (windows)	23,974 / 3,425 / 6,850

Positioning. We adopt Graph WaveNet as our representative GNN. It is accurate, convolutional (so it trains on modest hardware) and lets us ablate both the predefined and the learned graph. We pair it with strong, leakage-free feature baselines and add significance testing, which the literature rarely reports.

III. Dataset and Exploratory Analysis

A. METR-LA

METR-LA [6], [26] contains 5-minute average speeds from N=207 loop detectors on Los Angeles County freeways, from 1 March to 27 June 2012 (T=34,272 steps; Table I). It ships with sensor coordinates and 295,374 directed road-network distances, of which 11,753 are between sensors in the dataset. The HDF5 file was written by pandas 0.15, so we read its arrays directly with `h5py`.

B. Data quality

Missing observations are encoded as 0 mph, which is physically implausible on a freeway. They make up 8.1 % of readings. Fig. 1 shows two mechanisms: network-wide outages (vertical stripes, e.g. 26–27 March, when >50 % of readings are missing) and long single-sensor failures (the median sensor's longest gap is 381 steps, about 32 h). Speeds are capped at 70 mph by the sensors' processing. We therefore (i) mask zero targets out of every loss and metric, (ii) never impute labels, and (iii) use only *causal* forward-filling (last observation carried forward) when building features. Interpolation would use future values.

C. Validation and cleaning

`scripts/clean_data.py` first runs 12 integrity checks on the raw files (monotone 5-minute grid, sensor order, value range, coordinates, distances); all pass. It then assigns each reading at most one issue code: *network outage* (all 207 sensors read 0; 2,148 steps, 6.27 % of readings), *missing* (a single sensor reads 0; 1.84 %), *stuck* (an identical reading below the 70 mph cap repeated for ≥1 h; 274 readings, e.g. 64 mph for 7 h) and *spike* (Hampel filter, window 7, $|x - \tilde{x}| > 5 \cdot 1.4826\,\mathrm{MAD}$ and >20 mph; 6,142 readings, 0.09 %). Constant runs at the cap are night-time free flow and are kept, as are sustained low speeds, which are real congestion. Flagged readings are imputed by linear interpolation for gaps of up to 1 h and by the sensor's weekly profile (fitted on training data) for longer gaps. We validated this rule by hiding 2,000 random blocks of valid readings per gap length in the validation and test period (Fig. 2). Interpolation is best up to 1 h (MAE 2.2–3.1 mph), while beyond 3 h the profile is clearly better (4.0 vs. 8.6 mph at 24 h). The cleaned series serves analysis and the demo app. The models still read the raw data with zero targets masked, because interpolation uses future values and imputed targets would make our scores incomparable with published results.

D. Temporal structure

Weekday speeds show a pronounced evening rush hour (network mean 45.2 mph at 17:30) and a weaker morning dip, while weekends are flat (minimum 56.8 mph; Fig. 3). The autocorrelation function (Fig. 4) is 0.86 at lag 5 min and 0.57 at 1 h. It peaks every 24 h, with the 168 h peak (0.40) above the 24 h peak (0.36), so the seasonality is weekly. Subtracting each sensor's weekly profile (fitted on training data) removes the daily peaks: the 24 h ACF falls from 0.36 to 0.07. The residual *anomaly* is a short-memory process (ACF 0.41 at 1 h). This additive decomposition, speed = weekly profile + anomaly, motivates several features. Augmented Dickey–Fuller tests [27] reject a unit root for all ten sampled sensors ($p < 10^{-24}$), so the series need no differencing; their non-stationarity is seasonal, not a stochastic trend.

E. Spatial structure

Pairwise correlation decays with road distance (Fig. 5), and the decay persists after de-seasonalising, so it is not just a shared daily cycle. Graph-connected sensors have a mean anomaly correlation of 0.27, against 0.07 for pairs with no recorded road path. The Spearman correlation between distance and anomaly correlation is −0.20. The distance graph is strongly directed: 73 % of edges have no reverse edge, and five sensors have no outgoing edge. This justifies both direction-aware spatial features and a learned adjacency.

F. Split and distribution shift

Following [6], [8], windows are split chronologically 70/10/20 (train to 23 May, validation to 4 June, test to 27 June). Mean speed is stable across splits (58.6 / 58.9 / 57.8 mph), but the test period has more missing data (12.1 % vs. 7.1 % in training). This mild, realistic shift motivates the robustness analysis in Section VII.

IV. Feature Engineering

The tabular models (Ridge and gradient boosting) predict $y_{t+h,i}$, the speed of sensor i at horizon h, from features of the anchor time t. Each (t, i) pair is one row. Let $\tilde{x}$ be the causally forward-filled speed, $\tau(t) \in \{0, \ldots, 2015\}$ the time-of-week slot, and $\mu_i(\tau)$ the *weekly profile*: sensor i's mean observed training speed in slot τ. Table II lists the 34 features. Three design rules apply throughout:

1) **Causality.** Features use only $\tilde{x}_{\le t}$ and statistics fitted on the training period. Calendar features describe the *target* time $t+h$, which is known in advance. A unit test overwrites every value after t and checks that no feature changes.

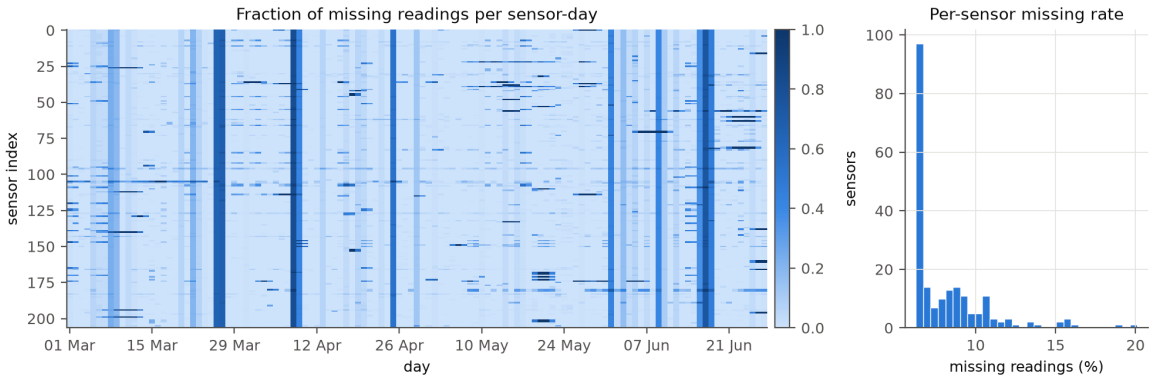


Fig. 1. Missingness. Left: fraction of zero readings per sensor-day. Vertical stripes are network outages; horizontal streaks are sensor failures. Right: distribution of per-sensor missing rates.

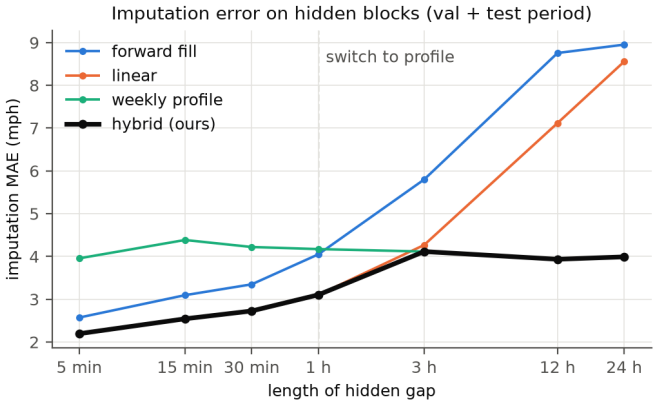


Fig. 2. Imputation error on artificially hidden gaps. The hybrid rule (linear for gaps $\leq$1 h, weekly profile otherwise) is the best method at every length.

2) **Decomposition.** Following the EDA, speed is split into the expected level $\mu_i(\tau(t+h))$ and the current anomaly $a_{t,i} = \tilde{x}_{t,i} - \mu_i(\tau(t))$. The model mainly has to learn how the anomaly decays.
3) **Direction-aware space.** Using the random-walk matrices $P_f = D_O^{-1}W$ and $P_b = D_I^{-1}W^\top$ (self-loops removed), we aggregate the speed, anomaly and 15-min trend of downstream (P_f) and upstream (P_b) neighbours. Congestion waves travel upstream, so downstream slowdowns give early warning.

Evaluation of features. We measure each group's contribution with a *cumulative ablation*: gradient boosting is retrained as groups are added and scored by validation MAE at 15/30/60 min. Individual features are assessed by *permutation importance* (the increase in validation MAE when one column is shuffled) on 60,000 held-out rows. Permutation importance avoids the high-cardinality bias of impurity-based importance. Results are in Section VII.

TABLE II
ENGINEERED FEATURES (34 IN SIX GROUPS).

Group	Features
Lags (12)	$\tilde{x}_{t-k,i}$, $k = 0, \ldots, 11$
Window (7)	Δ_1, Δ_3 differences; rolling mean, std, min over 1 h; #missing in window; missing-now flag
Calendar (4)	sin, cos of time-of-day of $t{+}h$; day of week; weekend flag
Profile (3)	$\mu_i(\tau(t{+}h))$; $\mu_i(\tau(t))$; anomaly $a_{t,i}$
Spatial (5)	$(P_f\tilde{x}_t)_i$, $(P_b\tilde{x}_t)_i$, $(P_f a_t)_i$, $(P_b a_t)_i$, $(P_f(\tilde{x}_t - \tilde{x}_{t-3}))_i$
Sensor (3)	training mean, std and missing rate of sensor i

V. Methodology

A. Problem formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$ be a directed graph with $N = |\mathcal{V}|$ sensors and weighted adjacency $W \in \mathbb{R}^{N\times N}$, and let $X_t \in \mathbb{R}^{N\times C}$ hold the C input signals at time t (speed and time of day). We learn

$$f_\theta : \big(X_{t-T'+1}, \ldots, X_t; \mathcal{G}\big) \mapsto \big(\hat{Y}_{t+1}, \ldots, \hat{Y}_{t+T}\big), \quad (1)$$

with $T' = T = 12$ (one hour), by minimising the masked mean absolute error

$$\mathcal{L}(\theta) = \frac{\sum_{h,i} m_{t+h,i}\,\big|\hat{y}_{t+h,i} - y_{t+h,i}\big|}{\sum_{h,i} m_{t+h,i}}, \qquad m = \mathbb{1}[y > 0]. \quad (2)$$

The mask removes missing targets. MAE's population minimiser is the conditional *median*, which suits the heavy left tail of speeds (sudden congestion): an MSE loss would let rare large errors dominate.

B. Graph construction

Following [6], $W_{ij} = \exp(-d_{ij}^2/\sigma^2)$ if this is $\geq \kappa = 0.1$ and 0 otherwise, where d_{ij} is the road distance from i to j and σ the standard deviation of distances. The Gaussian kernel turns distance into similarity, and the threshold keeps the graph sparse (1,722 non-zeros, matching the original preprocessing exactly). Because $d_{ij} \neq d_{ji}$ in general, W is asymmetric.

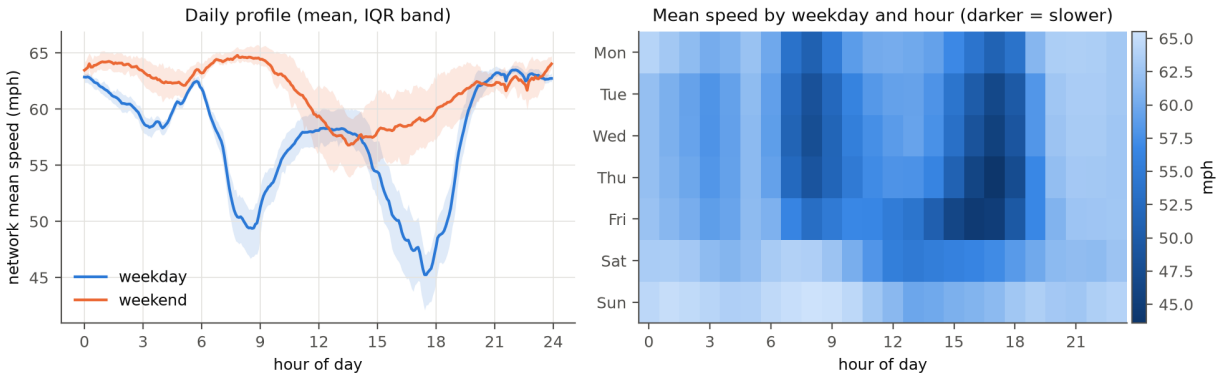


Fig. 3. Left: network-mean speed by time of day (line: mean; band: inter-quartile range across days). Right: mean speed by weekday and hour.

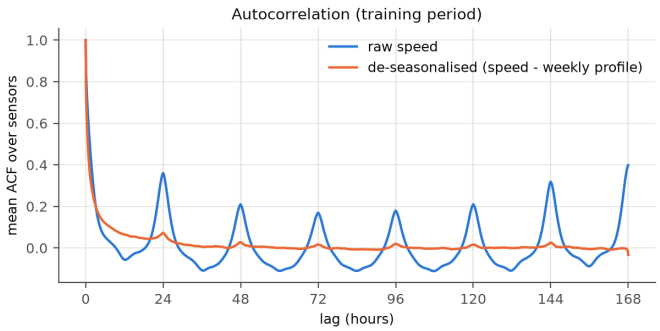


Fig. 4. Mean autocorrelation over sensors, for raw speeds and after removing the weekly profile.

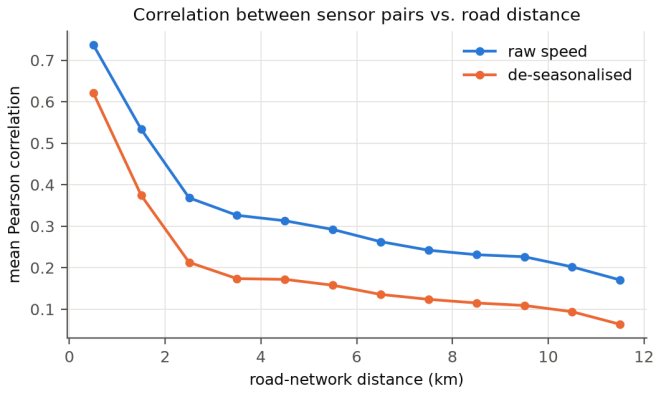


Fig. 5. Mean Pearson correlation between sensor pairs, binned by road distance.

C. Reference models

Historical Average (HA) predicts $\mu_i(\tau(t+h))$, the sample estimate of $\mathbb{E}[x \mid \text{sensor, time-of-week}]$. **Persistence** predicts $\tilde{x}_{t,i}$ for every horizon. It is the optimal forecast for a random walk (martingale), and its error grows with h because speed is mean-reverting.

D. Ridge regression

For each horizon, $\hat{\beta} = \arg\min_\beta \|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2 = (X^\top X + \lambda I)^{-1} X^\top y$ on standardised features [28]. The lag features are highly collinear (lag-1 ACF 0.86), so $X^\top X$ is ill-conditioned. The ℓ_2 penalty shrinks along low-variance eigen-directions, trading a little bias for a large variance reduction. λ is chosen on validation.

E. Gradient-boosted trees

Gradient boosting [11] builds an additive model $F_M(\mathbf{x}) = F_0 + \sum_{m=1}^{M} \nu\, h_m(\mathbf{x})$. Each regression tree h_m is fit to the pseudo-residuals $r_m = -\partial \ell(y, F)/\partial F|_{F_{m-1}}$. For the absolute loss $\ell = |y - F|$, $r_m = \text{sign}(y - F_{m-1})$, and the leaf values are set to the median of the residuals in each leaf, which makes the fit robust to outliers. We use the histogram variant [12], [29]: features are binned into ≤ 255 quantile bins, so split search costs $O(\text{bins})$ per feature rather than $O(n \log n)$. Shrinkage ν, leaf count, minimum leaf size, an ℓ_2 leaf penalty and early stopping control the bias–variance trade-off.

Direct multi-horizon strategy. We fit an independent model for every $h \in \{1, \ldots, 12\}$ rather than iterating a one-step model. Recursive forecasting feeds predictions back as inputs, so errors accumulate; the direct strategy avoids this and lets each horizon weight features differently [30].

F. LSTM

A two-layer LSTM [14] with 64 hidden units is shared across sensors. Each sensor's 12-step history (x_t, tod_t) is encoded by

$$[\mathbf{i}; \mathbf{f}; \mathbf{o}] = \sigma(W[\mathbf{h}_{s-1}; \mathbf{x}_s] + \mathbf{b}), \quad \mathbf{g} = \tanh(W_g[\mathbf{h}_{s-1}; \mathbf{x}_s] + \mathbf{b}_g),$$
$$\mathbf{c}_s = \mathbf{f} \odot \mathbf{c}_{s-1} + \mathbf{i} \odot \mathbf{g}, \qquad \mathbf{h}_s = \mathbf{o} \odot \tanh(\mathbf{c}_s). \tag{3}$$

An MLP maps the final $\mathbf{h}$ to all 12 outputs. The additive cell update keeps gradients from vanishing. Because weights are shared and no information crosses sensors, the LSTM isolates the value of *temporal* learning without spatial information.

G. Graph WaveNet

Graph WaveNet [8] stacks $L = 8$ layers (4 blocks $\times$ 2). Each layer combines a gated temporal convolution with a graph convolution, plus a residual path and a skip connection to the output head.

Gated dilated causal convolution. For input $\mathcal{X} \in \mathbb{R}^{N\times D\times S}$,

$$\mathcal{H} = \tanh(\Theta_1 \star_d \mathcal{X} + \mathbf{b}) \odot \sigma(\Theta_2 \star_d \mathcal{X} + \mathbf{c}), \tag{4}$$

where $\star_d$ is a causal 1-D convolution with kernel $K{=}2$ and dilation d. The sigmoid gate controls information flow, like an LSTM gate, but all time steps are computed in parallel. With dilations $1, 2$ repeated over four blocks, the receptive field is $1 + 4 \cdot (K{-}1)(1 + 2) = 13 \geq T'$, so every output sees the whole input hour.

Diffusion graph convolution. Traffic is modelled as a bidirectional random walk on $\mathcal{G}$, with forward and backward transition matrices $P_f = D_O^{-1}W$ and $P_b = D_I^{-1}W^\top$. With the support set $\mathcal{P} = \{P_f, P_b, \tilde{A}\}$,

$$Z = \mathcal{H}\,\Theta_0 + \sum_{P\in\mathcal{P}} \sum_{k=1}^{K_d} \big(P^\top\big)^k \mathcal{H}\,\Theta_{P,k}, \qquad K_d = 2, \tag{5}$$

so node j receives $\sum_i (P^k)_{ij}\mathcal{H}_i$, the mass a k-step random walk carries into j. P_f propagates information along the direction of travel and P_b against it, covering up to K_d hops at a cost of $O(K_d|\mathcal{E}|D)$ with sparse matrices [6]. The concatenated terms are mixed by a $1{\times}1$ convolution, which implements the Θ's.

Self-adaptive adjacency. The road graph is incomplete: some sensors have no outgoing edge, and dependencies such as parallel freeways are not captured by distance. Graph WaveNet therefore learns

$$\tilde{A} = \mathrm{softmax}\big(\mathrm{ReLU}(E_1 E_2)\big), \qquad E_1 \in \mathbb{R}^{N\times 10},\ E_2 \in \mathbb{R}^{10\times N}, \tag{6}$$

end-to-end. ReLU prunes weak links, and the row softmax makes $\tilde{A}$ a valid transition matrix. Its rank is ≤ 10, which regularises the $O(N^2)$ parameterisation.

Output. Skip connections from all layers are summed, passed through two $1{\times}1$ convolutions with ReLU, and mapped to 12 horizons at once (direct strategy). The model has 300,952 parameters and is trained with Eq. (2).

H. Statistical comparison

Diebold–Mariano test. For models A and B at horizon h, let $d_s = \overline{|e_s^A|} - \overline{|e_s^B|}$ be the loss differential of test window s, averaged over observed sensors. Under $H_0 : \mathbb{E}[d_s] = 0$,

$$\mathrm{DM} = \bar{d}\,\big/\sqrt{\hat{\sigma}^2_{\mathrm{LR}}/n} \ \xrightarrow{d}\ \mathcal{N}(0,1), \tag{7}$$

where $\hat{\sigma}^2_{\mathrm{LR}}$ is a Newey–West long-run variance with Bartlett weights and h lags [9], [31]. This accounts for the autocorrelation of overlapping forecast windows. **Block bootstrap.** 95 % confidence intervals for overall MAE resample whole test days with replacement [32], which preserves within-day dependence.

TABLE III
HYPER-PARAMETERS (SELECTED ON VALIDATION WHERE A RANGE IS SHOWN).

Model	Settings
Ridge	$\lambda \in \{10^{-2}, 1, 10^2, 10^3\}$, standardised features, one model per horizon
Grad. boosting	absolute loss; $(\nu, \text{leaves}) \in \{(0.1, 31), (0.1, 127), (0.05, 63)\}$; min leaf 100; $\ell_2{=}1$; $\leq$400 (800 for $\nu{=}0.05$) iterations, early stopping (patience 20); one model per horizon
LSTM	2 layers, 64 units, dropout 0.1, MLP head; Adam lr 10^{-3}, weight decay 10^{-4}, batch 32, $\leq$60 epochs, patience 10
Graph WaveNet	4 blocks $\times$ 2 layers, kernel 2, channels 32/32/256/512 (residual/dilation/skip/end), diffusion order 2, embedding dim 10, dropout 0.3; Adam lr 10^{-3}, weight decay 10^{-4}, clip 5, batch 32, $\leq$35 epochs, patience 8
GWN ablation	as above, exactly 8 epochs for every variant

VI. Experimental Setup

Protocol. We follow the standard METR-LA protocol [6], [8]: 12 input steps predict 12 output steps; windows are split chronologically 70/10/20 (23,974 / 3,425 / 6,850); and MAE, RMSE and MAPE are computed on raw mph over *non-zero* targets at horizons 3, 6 and 12 (15, 30 and 60 min). The scaler, weekly profile and sensor statistics are fit on the training period only. The test split is used once, after all model selection.

Inputs. Deep models receive two channels per sensor and step: z-scored speed (mean 58.58, std 12.83, from observed training values) and time of day in $[0, 1)$. Following the reference pipeline, missing inputs keep their raw value 0, which z-scores to -4.57; this is a distinctive value the network can learn to recognise. Tabular models receive the 34 features of Section IV.

Model selection and training. Table III lists the settings. Consecutive windows share 11 of 12 steps, so the tabular models use every fourth training window ($\approx$1.15 M observed rows per horizon) with little loss of information. The validation windows are split by parity: even windows select hyperparameters, odd windows drive early stopping. Deep models use Adam [33] with gradient-norm clipping and keep the checkpoint with the best validation MAE. Graph WaveNet uses the published architecture; we use batch size 32 (instead of 64) to fit the 8 GB unified memory of an Apple M3, and train for at most 35 epochs (instead of 100) because of compute limits. All code uses PyTorch [34] and scikit-learn [29], with seed 0.

VII. Results

Pending.

VIII. Conclusion

Pending.

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